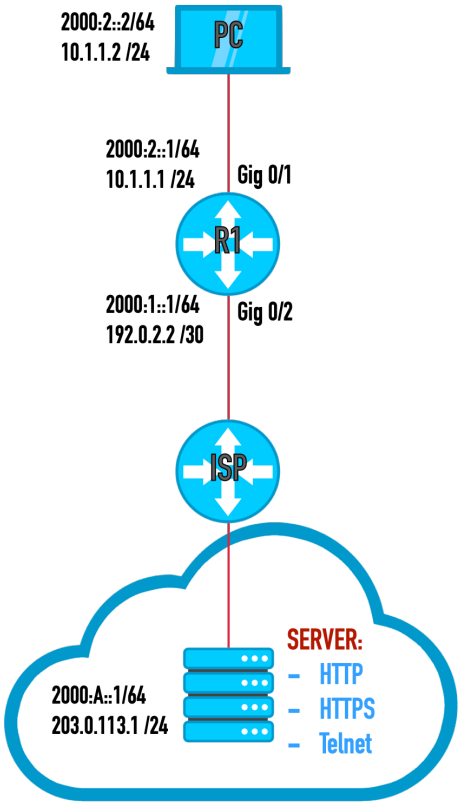
Control Plane Policing (CoPP)

Topology



Initial Configuration Commands

PC:

```
enable
conf t
host PC
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
```

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```
line con 0
logging synchronous
exit
no ip domain-lookup
no ip routing
ipv6 unicast-routing
int gig 0/1
ip address 10.1.1.2 255.255.255.0
ipv6 address 2000:2::2/64
no shut
ip default-gateway 192.0.2.1
ipv6 route ::/0 2000:2::1
end
copy run star
```

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
transport input telnet
line con 0
logging synchronous
exit
no ip domain-lookup
ipv6 unicast-routing
int gig 0/1
ip address 10.1.1.1 255.255.255.0
ipv6 address 2000:2::1/64
no shut
int gig 0/2
ip address 192.0.2.2 255.255.255.252
ipv6 address 2000:1::1/64
no shut
exit
ipv6 router rip ROUTE
int gig 0/1
```

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```
ipv6 rip ROUTE enable
exit
int gig 0/2
ipv6 rip ROUTE enable
exit
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
end
copy run star
```

ISP: (Will have a hostname of SERVER)

```
enable
conf t
host SERVER
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
transport input telnet
line con 0
logging synchronous
exit
no ip domain-lookup
ipv6 unicast-routing
int gig 0/1
ip address 192.0.2.1 255.255.255.252
ipv6 address 2000:1::2/64
no shut
int lo0
ip address 203.0.113.1 255.255.255.0
ipv6 address 2000:A::1/64
exit
ipv6 router rip ROUTE
int gig 0/1
ipv6 rip ROUTE enable
exit
int lo0
ipv6 rip ROUTE enable
exit
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
```

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```
exit
ip http server
ip http secure-server
end
copy run star
```

Lab Tasks

1. Let's start a continuous ping from our PC to R1's IP address of 10.1.1.1.
2. On R1, let's create an extended ACL with a number of 100. Also, let's permit ICMP messages from anywhere to anywhere.
3. Let's create a class map and let's name it PING. Match it to our access-group of 100.
4. Let's create a policy map and name it PING-POLICY.
5. Now that we are under Policy Map configuration mode, let's reference the previously created Class Map (PING) here.
6. Now we can identify the options that we want taken against the identified traffic. Let's police a target bit rate of 8000 bits per second, and then choose conform-action.
7. If our traffic is at or below 8000 bits per second, set the action to "transmit". If our traffic exceeds 8000 bits per second, set the action to "drop".
8. For the final step, we need to apply the service policy to the control plane interface. Select the "input" option to apply the policy to the control plane interface and reference the policy map name we created, PING-POLICY.
9. Go back to PC1 and see if the policing rate limits that we set up are affecting our continuous ping.
10. On R1, let's run the verification command "show policy-map control-plane" to see information about the policy map that is applied to our control plane.

Solution

Step 1: Let's start a continuous ping from our PC to R1's IP address of 10.1.1.1.

```
PC>en
PC#ping 10.1.1.1 repeat 1000000
Type escape sequence to abort.
Sending 1000000, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
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```

(##We can see here that we are completely successful and that we have no Control Plane Policing policies in place that are limiting our ICMP traffic.)

Step 2: On R1, let's create an extended ACL with a number of 100. Also, let's permit ICMP messages from anywhere to anywhere.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip access-list extended 100
R1(config-ext-nacl)#permit icmp any any
R1(config-ext-nacl)#exit
```

(##We are using this ACL to identify the traffic for policing, rather than for filtering purposes.)

Step 3: Let's create a class map and let's name it PING. Match it to our access-group of 100.

```
R1(config)#class-map ?
  WORD       class-map name
  match-all  Logical-AND all matching statements under this classmap
  match-any  Logical-OR all matching statements under this classmap
  type       Configure CPL Class Map

R1(config)#class-map PING
R1(config-cmap)#?
Class-map configuration commands:
  description  Class-Map description
  exit         Exit from class-map configuration mode
  match        classification criteria
  no           Negate or set default values of a command
```

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```
R1(config-cmap)#match ?
  access-group         Access group
  any                  Any packets
  application          Application to match
  cac                  Call Admission Control
  class-map            Class map
  cos                  IEEE 802.1Q/ISL class of service/user priority values
  destination-address  Destination address
  discard-class        Discard behavior identifier
  dscp                 Match DSCP in IPv4 and IPv6 packets
  fr-de                Match on Frame-relay DE bit
  fr-dlci              Match on fr-dlci
  input-interface      Select an input interface to match
  ip                   IP specific values
  metadata             Metadata to match
  mpls                 Multi Protocol Label Switching specific values
  not                  Negate this match result
  packet               Layer 3 Packet length
  precedence           Match Precedence in IPv4 and IPv6 packets
  protocol             Protocol
  qos-group            Qos-group
  security-group       Security group
  service              Service Instance to match
  source-address       Source address
  vlan                 VLANs to match

R1(config-cmap)#match access-group 100
R1(config-cmap)#exit
```

Step 4: Let's create a policy map and name it PING-POLICY.

```
R1(config)#policy-map PING-POLICY
```

Step 5: Now that we are under Policy Map configuration mode, let's reference the previously created Class Map (PING) here.

```
R1(config-pmap)#?
Policy-map configuration commands:
  class        policy criteria
  description  Policy-Map description
  exit         Exit from policy-map configuration mode
  no           Negate or set default values of a command

R1(config-pmap)#class PING
```

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(##Under contextual help, you are going to notice that there is no "match" statement this time, instead we have the "class" option.)

Step 6: Now we can identify the options that we want taken against the identified traffic. Let's police a target bit rate of 8000 bits per second, and then choose conform-action.

```
R1(config-pmap-c)#?
Policy-map class configuration commands:
  admit            Admit the request for
  bandwidth        Bandwidth
  compression      Activate Compression
  drop             Drop all packets
  exit             Exit from class action configuration mode
  fair-queue       Enable Flow-based Fair Queuing in this Class
  flow             Flow subcommands
  log              Log IPv4 and ARP packets
  measure          Measure
  netflow-sampler  NetFlow action
  no               Negate or set default values of a command
  police           Police
  priority         Strict Scheduling Priority for this Class
  queue-limit      Queue Max Threshold for Tail Drop
  random-detect    Enable Random Early Detection as drop policy
  service-policy   Configure QoS Service Policy
  set              Set QoS values
  shape            Traffic Shaping

R1(config-pmap-c)#police ?
  <8000-2000000000>  Target Bit Rate (bIts per second) (postfix k, m, g
                     optional; decimal point allo
  cir                Committed information rate
  rate               Specify police rate, PCR for hierarchical policies or
SCR
                     for single-level ATM 4.0 policer policies

R1(config-pmap-c)#police 8000 ?
  <1000-512000000>  Burst bytes
  bc                Conform burst
  conform-action    action when rate is less than conform burst
  pir               Peak Information Rate
  <cr>              <cr>
R1(config-pmap-c)#police 8000 conform-action
```

(##Conform-action allows us to decide what action to take if our traffic matches or falls below the 8000 bit rate.)

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Step 7: If our traffic is at or below 8000 bits per second, set the action to "transmit". If our traffic exceeds 8000 bits per second, set the action to "drop".

```
R1(config-pmap-c)#police 8000 conform-action ?
  drop                              drop packet
  set-clp-transmit                  set atm clp and send it
  set-cos-inner-transmit            rewrite packet inner cos and send it
  set-cos-transmit                  rewrite packet cos and send it
  set-discard-class-transmit        set discard-class and send it
  set-dscp-transmit                 set dscp and send it
  set-dscp-tunnel-transmit          rewrite tunnel packet dscp and send it
  set-frde-transmit                 set FR DE and send it
  set-mpls-exp-imposition-transmit  set exp at tag imposition and send it
  set-mpls-exp-topmost-transmit     set exp on topmost label and send it
  set-prec-transmit                 rewrite packet precedence and send it
  set-prec-tunnel-transmit          rewrite tunnel packet precedence and send
                                    it
  set-qos-transmit                  set qos-group and send it
  transmit                          transmit packet

R1(config-pmap-c)#police 8000 conform-action transmit ?
  exceed-action  action when rate is within conform and conform + exceed
burst
  <cr>           <cr>

R1(config-pmap-c)#police 8000 conform-action transmit exceed-action ?
  drop                              drop packet
  set-clp-transmit                  set atm clp and send it
  set-cos-inner-transmit            set inner cos and send it
  set-cos-transmit                  set cos and send it
  set-discard-class-transmit        set discard-class and send it
  set-dscp-transmit                 set dscp and send it
  set-dscp-tunnel-transmit          rewrite tunnel packet dscp and send it
  set-frde-transmit                 set FR DE and send it
  set-mpls-exp-imposition-transmit  set exp at tag imposition and send it
  set-mpls-exp-topmost-transmit     set exp on topmost label and send it
  set-prec-transmit                 rewrite packet precedence and send it
  set-prec-tunnel-transmit          rewrite tunnel packet precedence and send
                                    it
  set-qos-transmit                  set qos-group and send it
  transmit                          transmit packet

R1(config-pmap-c)#police 8000 conform-action transmit exceed-action drop
R1(config-pmap-c-police)#end
```

(##What's happening here is that we're policing the ICMP traffic: if it arrives at a rate of less than 8000 bits per

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second, it will be transmitted; if it exceeds that rate, it will be dropped.)

Step 8: For the final step, we need to apply the service policy to the control plane interface. Select the "input" option to apply the policy to the control plane interface and reference the policy map name we created, PING-POLICY.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#control-plane
R1(config-cp)#?
Control Plane configuration commands:
  exit            Exit from control-plane configuration mode
  no              Negate or set default values of a command
  service-policy  Configure QOS Service Policy

R1(config-cp)#service-policy ?
  input   Assign policy-map to the input of an interface
  output  Assign policy-map to the output of an interface
  type    type of the policy-map

R1(config-cp)#service-policy input PING-POLICY
R1(config-cp)#end
```

Step 9: Go back to PC1 and see if the policing rate limits that we set up are affecting our continuous ping.

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```

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(##You will notice some gaps in there. We are now experiencing drops in ICMP replies. This is because of the policing rate limits that we set up.)

Step 10: On R1, let's run the verification command "show policy-map control-plane" to see information about the policy map that is applied to our control plane.

```
R1#show policy-map control-plane
 Control Plane

  Service-policy input: PING-POLICY

    Class-map: PING (match-all)
      375 packets, 42750 bytes
      5 minute offered rate 5000 bps, drop rate 0000 bps
      Match: access-group 100
      police:
          cir 8000 bps, bc 1500 bytes
        conformed 350 packets, 39900 bytes; actions:
          transmit
        exceeded 25 packets, 2850 bytes; actions:
          drop
        conformed 5000 bps, exceeded 0000 bps

    Class-map: class-default (match-any)
      9 packets, 1584 bytes
      5 minute offered rate 0000 bps, drop rate 0000 bps
      Match: any
R1#show policy-map control-plane
 Control Plane

  Service-policy input: PING-POLICY

    Class-map: PING (match-all)
      529 packets, 60306 bytes
      5 minute offered rate 6000 bps, drop rate 0000 bps
      Match: access-group 100
      police:
          cir 8000 bps, bc 1500 bytes
        conformed 494 packets, 56316 bytes; actions:
          transmit
```

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```
      exceeded 35 packets, 3990 bytes; actions:
        drop
      conformed 6000 bps, exceeded 0000 bps

  Class-map: class-default (match-any)
    12 packets, 1878 bytes
    5 minute offered rate 0000 bps, drop rate 0000 bps
    Match: any
```

(##Notice how we can see the number of packets that have been conformed to our policing. Also notice how the number of conformed packets went up the second time we ran that command.)
(##Notice how we can see the number of packets that have exceeded our policing. Also notice how the number of exceeded packets went up the second time we ran that command.)
(##These numbers are growing because of the continuous ping coming from the PC into R1. This policing policy is rate limiting the ICMP traffic that is arriving on our router.)

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